Chapter route

How can an OS control heterogeneous devices through standard interfaces while controlling latency, throughput, isolation, and errors?

Prerequisites are interrupts, DMA, memory mapping, blocks, and files.

- Trace an application request through device completion.
- Distinguish blocking, nonblocking, asynchronous, and buffered I/O.
- Locate bottlenecks beyond the CPU.

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Three I/O timelines

A read(fd, buf,4096) can hit page cache or start a device. The thread can be running, blocked, or ready; the request queued, in flight, or complete; and the buffer owned by CPU or device. These are separate state machines.

Assume one blocking thread, cache miss, and successful DMA. At t=0 the syscall starts; at 2 µs validation/submission finishes; at 3 the thread sleeps; at 102 DMA completes; at 104 the handler records completion and wakes the thread; at 110 scheduling runs it; at 112 it returns to user mode. Device service takes about100 µs, call latency112 µs, and wakeup-to-execution wait6 µs.

Before successful data is observed, it must be CPU-readable. Ready does not mean returned. Scheduling, notification, and visibility have separate synchronization conditions.

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One interface, several contracts

A descriptor denotes an opened object and access authority, not a hardware address. Regular files support seek; pipes/sockets normally do not. Terminals can apply line discipline; ioctl performs type-specific control. A common entry point does not enable every operation.

Distinguish acceptance into a buffer, driver submission, controller completion, media persistence, and remote application processing. These are not one universal linear sequence: a local file may reach durability under its filesystem/device synchronization contract, whereas network transmission and remote application processing have protocol-specific meanings. A write returning 4096 does not universally establish all five. Durability needs a contract supported by filesystem and device; transmission success does not prove a receiving application committed. Numerical cases here are explicit teaching models, not machine measurements.

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Ports, MMIO, controllers, and buses

A device is the peripheral; its controller decodes commands, retains status, and transfers data. CPUs access ports through dedicated instructions or MMIO registers through protected mappings. MMIO is not ordinary RAM; caching, combining, and reordering follow architecture/driver I/O-accessor and barrier rules.

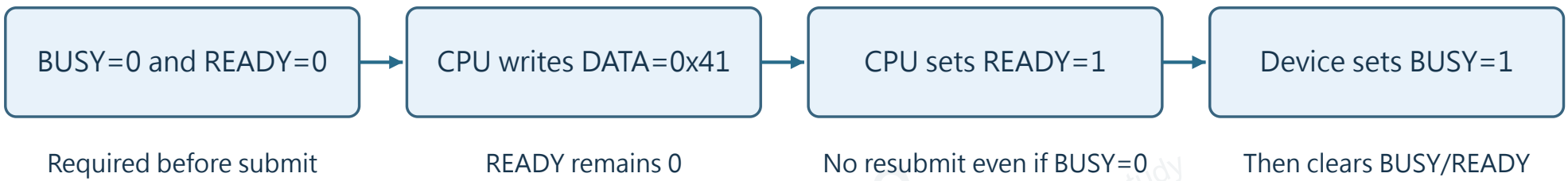
The teaching controller has STATUS, COMMAND, and DATA. STATUS bit0=BUSY, bit1=ERROR; COMMAND bit0=READY. Require BUSY=0 and READY=0, write DATA=0x41, select output, then publish READY. Publishing READY first can expose old data. Multiple CPU submitters additionally need a lock/queue; BUSY alone is not mutual exclusion.

A bus arbitrates shared transfers. Daisy-chain position/topology affects access and interrupt routing without removing permission checks.

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Polling handshake and invariant

PIO output 0x41: prepare data before publishing



can_submit = !BUSY && !READY. Do not overwrite DATA before completion.

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Polling handshake and invariant (continued)

Step	Actor	BUSY	READY	DATA	Next action
0	Initial	0	0	Old	Driver may submit
1	CPU writes data	0	0	0x41	Publish command
2	CPU sets READY	0	1	0x41	Controller accepts
3	Controller accepts	1	1	0x41	Output data
4	Completion clears state	0	0	0x41	CPU checks ERROR

can_submit=(BUSY==0 && READY==0). READY=1, BUSY=0 means published but not yet accepted. Even a single producer immediately attempting another byte must not overwrite this window. Testing only BUSY can replace0x41 with 0x42 before consumption.

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Polling handshake and invariant (continued)

The correct trace is (0,0, old)→publish(0,1,0x41)→reject another submit, retaining0x41→accept(1,1,0x41)→complete(0,0,0x41)→publish(0,1,0x42). Under one producer, both-bit checking, and required ordering, can_submit stays false from publication through completion, preserving DATA. A device that never completes defeats termination; timeout does not prove it stopped using buffers.

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Interrupts notify without moving every byte

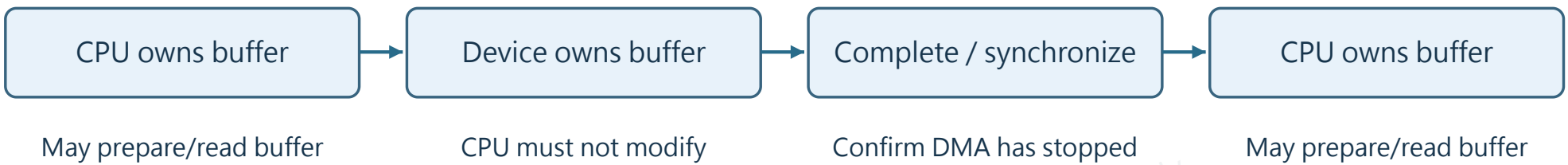
The controller raises an IRQ. At a permitted boundary the CPU saves required state and dispatches by vector. The handler checks source/status, acknowledges, updates the request, queues deferred work or wakes waiters, and returns. Shared IRQs require source identification rather than blindly completing the queue head.

Maskable interrupts, NMI, and synchronous exceptions differ; a low vector range is not uniformly NMI. Short handlers and sleepable later processing follow OS rules. An interrupt saves the required CPU control state and enters its handler; a privilege-mode transition occurs if it interrupted user mode, not necessarily when already in kernel mode. Switching threads additionally requires a scheduling context switch.

Lost wakeup occurs when a waiter reads done=false, IRQ sets true and wakes nobody, and the waiter subsequently sleeps. Under one lock, check the predicate, enqueue, and atomically release the lock while sleeping. Recheck in a while loop after waking; the IRQ publishes done under corresponding synchronization. The predicate stores completion evidence, while notification alone need not accumulate.

DMA addresses, coherence, and lifetime

4096 B receive: address mapping and ownership



Before submit: map/cache maintenance → descriptor/ownership → doorbell

After terminal: synchronize → publish → unmap/free only after hardware stops accessing

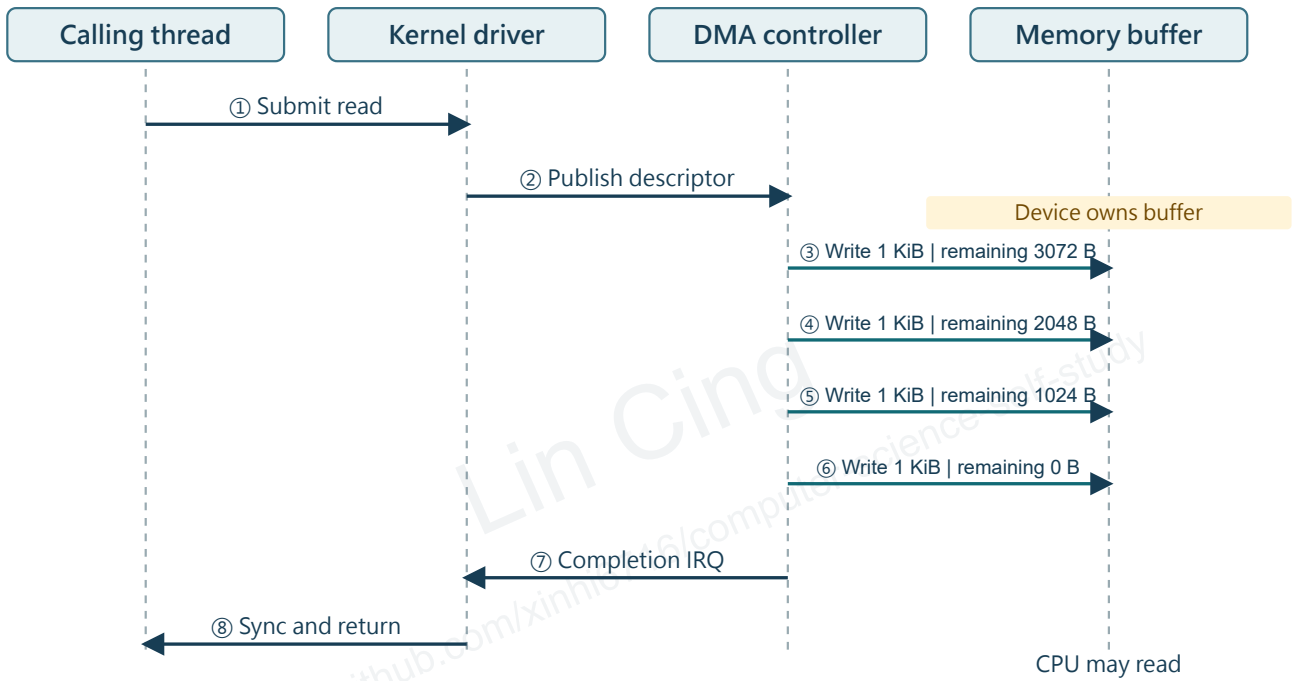
Prepare descriptor {device_addr, dma_addr, length=4096, direction=FROM_DEVICE}, mapping, and required cache maintenance before publication. Hardware transfers and notifies completion. IOMMU translation means a DMA address need not equal user VA or physical address. IOMMU confinement does not replace descriptor authorization.

DMA addresses, coherence, and lifetime (continued)

After handing the receive buffer to the device, CPU must not access it until completion and synchronization. Retain descriptor, mapping, and pages until hardware can no longer access them. Unmapping user VA does not cancel DMA. Noncoherent systems especially require the specified DMA API; a CPU barrier does not automatically invalidate caches.

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One DMA read: separate control and data paths



One DMA read: separate control and data paths (continued)

This example reads 4096 B in four 1024 B transfers and raises one completion IRQ; neither is a universal DMA rule. The driver maps and publishes a descriptor. The device owns the buffer during DMA; CPU use resumes only after completion and required synchronization. Waking a thread does not run it immediately.

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Transfer and notification costs

For 4 KiB, PIO moving 4 B per 80 ns needs 1024 operations, or 81.92 µs CPU time. DMA submission 2 µs plus IRQ/cleanup 1 µs costs 3 µs CPU time, while device transfer still takes 100 µs.

Periodic checks costing 100 ns each, every 5 µs, perform about 20 checks during service, assuming the CPU does other work or sleeps between checks, costing 2 µs CPU, with 0–5 µs detection delay, averaging 2.5 µs under uniform phase. At 2000 completions/s and 1 µs/IRQ, demand is 0.002 s/s=0.2% CPU. Periodic 5 µs checks need 200000 checks/s, or 2% CPU for the checks alone. Spinning continuously between checks would occupy nearly 100% of a core.

DMA still consumes memory/bus bandwidth. Overlapped CPU work does not remove system cost. Small transfers can favor PIO's lower setup cost. Compare fixed bytes, latency, and CPU denominators.

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Device classes, mapping, and timers

Block devices address blocks and can support random access. Character devices provide sequential data with device-specific record/seek rules. File mmap maps contents through faults/page cache; controller MMIO accesses registers.

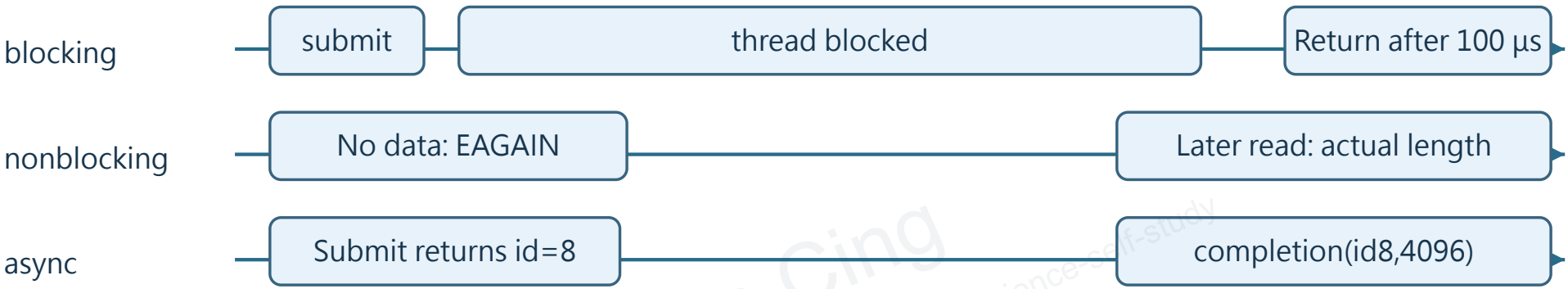
Raw/direct I/O bypasses selected services while retaining permission, alignment, lifetime, and isolation requirements. Sockets add endpoints/connections/transports. Streams do not preserve write boundaries; datagrams have message/truncation rules. Readiness transfers no bytes and does not prevent a competing thread consuming them first.

Timers provide clocks, interrupts, and deadlines. A deadline queue at now7 ms contains10,13,13. Program10, remove that expiration, program13, then remove both equal deadlines. A trigger deadline does not promise immediate callback execution.

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Blocking, nonblocking, and asynchronous control flow

One read, three API contracts



Device service time is unchanged; call control flow and notification timing differ.

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Blocking, nonblocking, and asynchronous control flow (continued)

Blocking can sleep and returns an ended operation, possibly short. For a positive-length nonblocking pipe or stream socket read, EAGAIN applies when no data is available, the writing side remains open, and the operation would otherwise wait; EOF can instead return 0. Regular-file O_NONBLOCK behavior depends on the object and system contract. A read after poll/select still handles EAGAIN if a competing reader consumes the data. Async submission and completion differ; completion carries ID, status, actual length. Failed submission may create no request.

A 512 B request with 200 B available can return 200. Then done=200, remaining=312, dst=buf+200. Another 112 gives done=312, remaining=200; subsequent zero means EOF. Deliver the short 312 B or report protocol truncation. Never add -1 to done. Handle EINTR/EAGAIN under the API and progress already made.

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Partial vector I/O

For writev lengths[100,200,300], successful250 B consumes100 then150. Retry [base1+150,50], [base2,300], totaling350 B. Reusing the original vector duplicates250 B.

Vectors reduce calls and copies into contiguous staging storage. They do not universally guarantee device zero-copy or arbitrary-length atomicity. Atomicity, order, and durability are independent. Delayed async buffers live through completion; a kernel copy changes cost/lifetime. Failed cancellation or an already-completed response does not authorize freeing memory hardware may still use.

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Queues, buffers, caches, and spools

Scheduling selects service order; buffering absorbs rate/unit mismatches and preserves copy semantics; caching retains reusable data; spooling queues complete jobs for relatively exclusive devices. One region can be buffer and cache, with distinct correctness conditions.

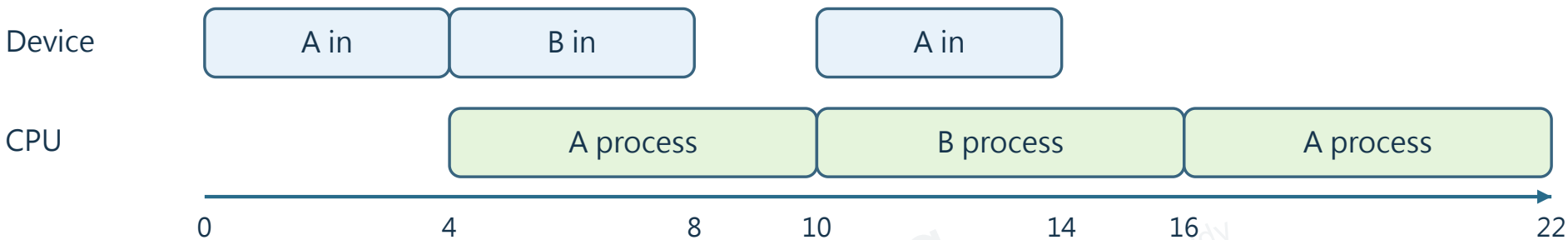
Reads arrive for 40,10,41. Reordering10→40→41 can reduce HDD movement but cannot cross dependencies/barriers. This mechanical-seek model is not SSD latency. Fairness/deadlines/throughput can conflict.

A dirty write-back cache can hold the only newest copy. Cache is not always an identical duplicate elsewhere. Accepted buffered write, device completion, and durability differ.

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Double buffering with occupied slots

Alternate two slots; hand only a free slot to the device



ms. Input 4 ms/block; CPU 6 ms/block. B waits at 8–10; A waits at 14–16.

Fill 0–4; overlap 4–16; drain 16–22. Three blocks finish at 22 ms, not 18 ms.

Input takes 4 ms/block and CPU6 ms. One buffer takes 30 ms for three blocks. With two, A inputs0–4 and processes4–10. B inputs4–8, waits8–10, processes10–16. A inputs10–14, waits14–16, processes16–22. Total22 ms; steady period=max(4,6)=6 ms.

At4–8 the units overlap, but at 8 hardware cannot overwrite A while CPU owns it. For N blocks with full overlap and two sufficient slots,

Double buffering with occupied slots (continued)

$$T = t_{in} + t_{cpu} + (N-1)\max(t_{in}, t_{cpu}).$$

Finite batches retain fill and drain time.

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Copy semantics, printing, and errors

An application writes ABC then changes its buffer to XYZ. If write copies/finishes consumption before returning, output remains ABC. Zero-copy async may forbid modification until completion. State the applicable contract.

The spooler persists three-page job J17 and prints. Page2 prints but acknowledgment is lost. Resending can duplicate pages; deleting can omit pages. A persistent host job ID prevents duplicate host job creation, not necessarily duplicate physical effects. Without device progress/deduplication, report outcome unknown.

Transient failures can permit retry, while non-idempotent operations cannot be blindly resent. Length/failure conventions determine valid errno use; stale errno does not make a successful call fail. Controller sense/error logs provide detail.

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Permissions and kernel objects

P fd4→process table→open-file object{offset 8192, readable=true, ops=regular_file_ops}→inode{logical 2→physical70}→block-driver queue. Some descriptors share an open offset; independent opens need not.

Validate fd/mode/range/user pointer, then VFS dispatches by type. Page-cache hit copies directly; miss enters block/driver. Socket ops receive network bytes rather than using block 70. Name resolution and read dispatch differ; read need not walk the pathname anew.

Privilege/mapping permissions protect ports, MMIO, and control registers. Validate/appropriately pin user buffers. Device authority does not authorize arbitrary-memory DMA. Reservations can deadlock when A holds scanner waiting printer while B holds printer waiting scanner. Ordered acquisition, joint allocation, or recovery must break the cycle.

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Complete address translation

Mount /data selects device(8,2); /data/log.bin has inode 37; open creates fd4. Read4096 B at offset 8192 gives logical 2, displacement0 for 4096 B blocks. Inode maps logical 2 to device block 70. With512 B sectors, partition LBA=560, count8. Major 8/minor 2 selects driver/partition; startLBA 2048 gives whole-disk LBA 2608. The controller additionally uses a DMA mapping. Fd, inode, user VA are not LBAs. This fixed teaching model does not describe every UNIX implementation identically.

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Address translation for one read

One read: from file offset to disk sectors

Block: 4,096 B · Sector: 512 B · Partition start: LBA 2,048

1	Open file	fd = 4; read offset = 8,192 B
2	File system	inode 37; logical block = 8,192 ÷ 4,096 = 2
3	Block mapping	logical block 2 → device block 70
4	Partition sector	70 × 8 sectors/block = LBA 560
5	Whole-disk sector	2,048 + 560 = LBA 2,608 (first sector)

A 4,096 B read spans 8 sectors: whole-disk LBA 2,608–2,615.

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Completion, timeout, and abort races

NEW→QUEUED→IN_FLIGHT→DONE or ABORTED publishes at most once; a device that never responds need not reach a terminal state. Timeout requests abort without proving ABORTED. Release mappings/pages only after normal completion or an abort acknowledgment confirming hardware quiescence, followed by the platform-required DMA synchronization.

Request 8 is submitted at t=0, times out at t=100, requests abort at t=101, completes at t=102, and receives an already-completed abort reply at t=103. A lock/CAS protects terminal state. Completion at t=102 wins; the abort reply at t=103 only releases bookkeeping without waking/unmapping twice. A timeout may be an early status notification, not terminal completion. If DMA targets a user buffer, the API must require the caller to retain it and not access or reuse it until final completion/quiescence; pinning pages alone cannot prevent reuse. Alternatively, DMA may use a kernel-owned bounce buffer so that the user buffer is no longer accessed by the device. Specify notification, terminal completion, and reclamation separately.

Completion, timeout, and abort races (continued)

Only a branch atomically changing nonterminal to terminal obtains publication authority; unmapping also requires hardware quiescence and the required synchronization. All others see terminal and refrain. Two winners would require a forbidden second terminal transition, a contradiction. This proves at-most-once completion, neither eventual response nor exactly-once external effects.

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Buffer and control-block lifetimes

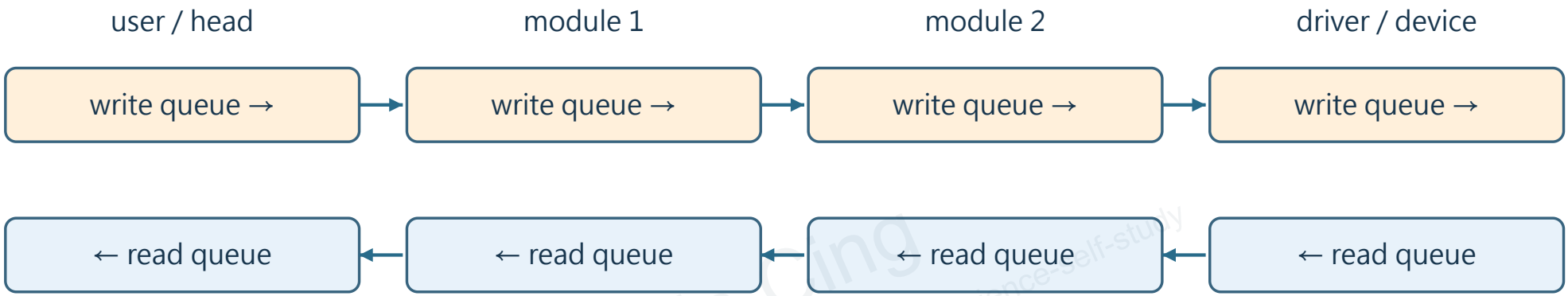
Stopped DMA permits buffer synchronization/unmapping, but timers/IRQs/abort callbacks may still read request state. The winner cannot immediately free its control block. Every queued/active callback already holds a stable reference and releases on exit. Canceling a timer cannot drop its reference without proving callback exit.

At101 owner, IRQ, abort hold count3. IRQ wins102, cleans DMA, exits3→2; owner delivers result and exits2→1; abort103 reads DONE, cleans bookkeeping, exits1→0, then free. An in-flight timer adds a reference. Acquisition itself requires lifetime protection; incrementing through a freed raw pointer is unsafe.

While any callback can read control state, a positive reference preserves it. Together with terminal checking this prevents double completion and use-after-free. Slot/ID reuse additionally requires generation or stable event identity against late events.

Bidirectional message queues

Historical STREAMS: two queue directions per component



Write moves right; read moves left. Full ordinary data queues propagate backpressure upstream.

Use historical System V/Solaris STREAMS. Head meets users, driver meets devices, ioctl pushes/pops modules. Every component has read/write queues; read moves toward head, write toward driver. Data/control messages do not require every module to copy payload.

Bidirectional message queues (continued)

Put procedures can process/forward immediately or enqueue. Service procedures drain runnable queues. Read/write expose byte streams; getmsg/putmsg can preserve control/data portions. Blocking at the user head does not mean synchronous waiting through every module.

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Numerical backpressure

Q capacity 1000 B, high 800, low 300, messages 200. Starting600, one accepted message gives800 and FULL, stopping upstream. Service yields600,400,200; below300 back-enables upstream. Hysteresis avoids switching repeatedly at one threshold.

Two upstream messages still consume memory. Full upstream queues eventually block the head writer. An unpausable hardware source may drop when all fill; IRQ cannot sleep indefinitely. Actual STREAMS priority/accounting is richer. This model covers ordinary fixed200 B data messages only.

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Resource throughput limits

Requests4 KiB; device100000/s; CPU5 µs/request gives200000/s on one core. Memory1 GiB/s with 12 KiB traffic/request, 4 KiB DMA plus copy read 4/write 4, gives87381.33/s. Thus overall throughput≤min(100000,200000,87381.33)=87381.33/s, a bound rather than achieved performance.

Truly removing that copy gives4 KiB traffic and262144/s memory bound; device100000/s becomes limiting. Count any remaining traffic. More cores cannot overcome the original memory bound.

Batching eight arrivals10 µs apart adds70 µs for first,0 for last, average35 µs. At1 µs/IRQ, CPU cost/request falls from 1 to0.125 µs at a latency cost.

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Latency and Little's relation

For finite steady-state means over one boundary, $L = \lambda W$. At80000/s and250 µs queue-plus-service response, L=20. With100 µs service, in-service count8, queue12, waiting150 µs. Queue count must not be paired with total response.

Minimum100 µs latency and100000/s throughput require average10 in flight, but queue depth10 does not guarantee that rate. Beyond saturation, more queueing can raise W without raising λ. At110000 arrivals/s and at most100000 completions/s without dropping/throttling, queue grows and finite steady-state L is unavailable.

This explicit conservation-model extension assumes neither exponential service nor the M/M/1 formula $W = 1/(\mu - \lambda)$. The latter needs additional distribution assumptions.

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One read across layers

Fd4, offset 8192→inode 37 logical 2→partitionLBA 560→whole-diskLBA2608→DMA4096 B. Thread blocked, request in flight, device owns buffer. Completion establishes CPU visibility, publishes state, readies thread; scheduling later returns it.

Cache hits skip lower layers. Nonblocking can return EAGAIN without bytes; async can return submission first. Short reads, errors, permissions, and lifetime remain. Check pre/postconditions along naming/authorization→queue→device→publication→use→reclamation rather than infer durable effects from one callback.

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Costs and observable guarantees

PIO/DMA choose transfer, poll/IRQ detection, blocking/async call contract. These can combine. Buffer/cache/spool solve different problems; copies can improve performance while separating return from durability.

State bytes/request, CPU time/request, bus bytes/request, service time, and observation boundary. Compare equal denominators; retain finite fill/drain and steady-state stability. Exercises retain intermediate states and general arguments rather than equating a few traces with proof.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 13.

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